7	**(a)**		$\frac{x^6}{8} \geq 0$ or $x^6 \geq 0$	**B1** **[1]**	**2.4**	Do not accept $\frac{x^6}{8}$ is always positive (ie > 0)
7	**(b)**		The expansion with two terms is a better approximation than the one with three terms but it should be the other way round.	**E1** **[1]**	**2.3**	O.E. See exemplars
7	**(c)**		$1+\frac{1}{2}\left(-x^3\right)+\frac{1}{2}\left(-\frac{1}{2}\right)\frac{\left(-x^3\right)^2}{2!}+\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\frac{\left(-x^3\right)^3}{3!}+\ldots$ $1-\frac{x^3}{2}-\frac{x^6}{8}-\frac{x^9}{16}\ldots$	**M1** **A2** **[3]**	**1.1a** **1.1** **1.1**	Working for 3rd or 4th term correct Must see the negative x^3 A2 all terms correct or A1 for three terms correct
7	**(d)**		$\lvert x\rvert < 1$ oe	**B1** **[1]**	**1.1**	
7	**(e)**		Goes through 2.5 on *y*-axis and (1,0) Right shape for values of *x* from –1 to 1	**B1** **B1** **[2]**	**2.2a** **1.1**	Curve reaches between (-1, 3) to (-1, 4.5)
7	**(f)**		**DR** $2.5\int_{-0.75}^{0.75}\left(1-\frac{x^3}{2}-\frac{x^6}{8}-\frac{x^9}{16}\right)\mathrm{d}x$ $[2.5]\left[x-\frac{x^4}{8}-\frac{x^7}{56}-\frac{x^{10}}{160}\right]_{-0.75}^{0.75}$	**M1** **M1**	**3.1b** **1.1**	For their expression from 7(c) in an integration. Condone d*x* missing. 2.5 and limits needed but may be seen later For integrating their expression (3 terms or more) allow one error Limits could be wrong or missing here

			$[2.5]\left[\left(0.75-\frac{0.75^4}{8}-\frac{0.75^7}{56}-\frac{0.75^{10}}{160}\right)-\left(-0.75-\frac{(-0.75)^4}{8}-\frac{(-0.75)^7}{56}-\frac{(-0.75)^{10}}{160}\right)\right]$	**M1**	**1.1**	For attempt at their limits substituted into their integrand. Substitution *must* be seen.
			$3.74\ m^2$	**A1**	**3.2a**	AWRT 3.74 from correct working. Must include units. 1.495 will probably get either **M2** or **M3** Annotate final page
			Special Case If trapezium rule used on original function allow **M1 A1** maximum			Likely answers If 2 strips used: $3.71m^2$ If 3 strips used: $3.72\ m^2$ If 6 strips used: $3.73\ m^2$
				[4]		

Exemplars for 7b

Accept (eg)

The 3-term expansion is further away than the one with 2 terms

The 3-term expansion moves away from the 2-term expansion

The 3-term expansion is on the wrong side of the graph of the 2-term expansion

The 3-term expansion goes up at the end

The 3-term expansion should be more accurate than a 2-term expansion [and it isn't]

The 3-term expansion moves away from the other 2

Do not accept (eg)

They don't follow the same shape